Week 4 — Credit Default Classification Report

1. Problem and data

The goal is to predict if a customer will default on a loan. The synthetic dataset has 4,000 rows and 8 unnamed features. We use 3,000 rows for training and 1,000 for testing; the test set has 73 defaults (7.3%), so the classes are imbalanced.

2. Model and training

I used StandardScaler and logistic regression with C = 1.0 and max_iter = 1000. I compared several probability thresholds instead of assuming 0.50 is best. A lower threshold can catch more defaults.

3. Results

Threshold	Precision	Recall	F1	ROC-AUC
0.50	0.919	0.466	0.618	0.871
0.35	0.830	0.603	0.698	0.871
0.25	0.701	0.644	0.671	0.871
0.15	0.486	0.712	0.578	0.871

Test ROC-AUC is 0.871, which measures how well the model ranks defaults above non-defaults. At threshold 0.15, recall is 71.2%, so it finds about 71% of real defaults. Precision is the share of flagged customers who really default; recall is the share of real defaults that are found.

4. Decision and cost

A missed default (FN) costs 10 units and a false alarm (FP) costs 1 unit. Cost = 10 × FN + FP. The lowest tested cost is 265 units at threshold 0.15. At threshold 0.50, cost is 393 units. I choose 0.15 because it gives the lowest cost under these assumptions.

5. Limitations

The 8 features have no names, so we cannot explain their business meaning. The data is synthetic, so the results may not work on real bank customers. Also, the assumed error costs may differ from real costs.

6. Next step

Test class_weight="balanced" or SMOTE and compare precision-recall AUC. Validate the chosen threshold on new data before using the model.

github repository: https://github.com/tomley80/W4_machine_learning